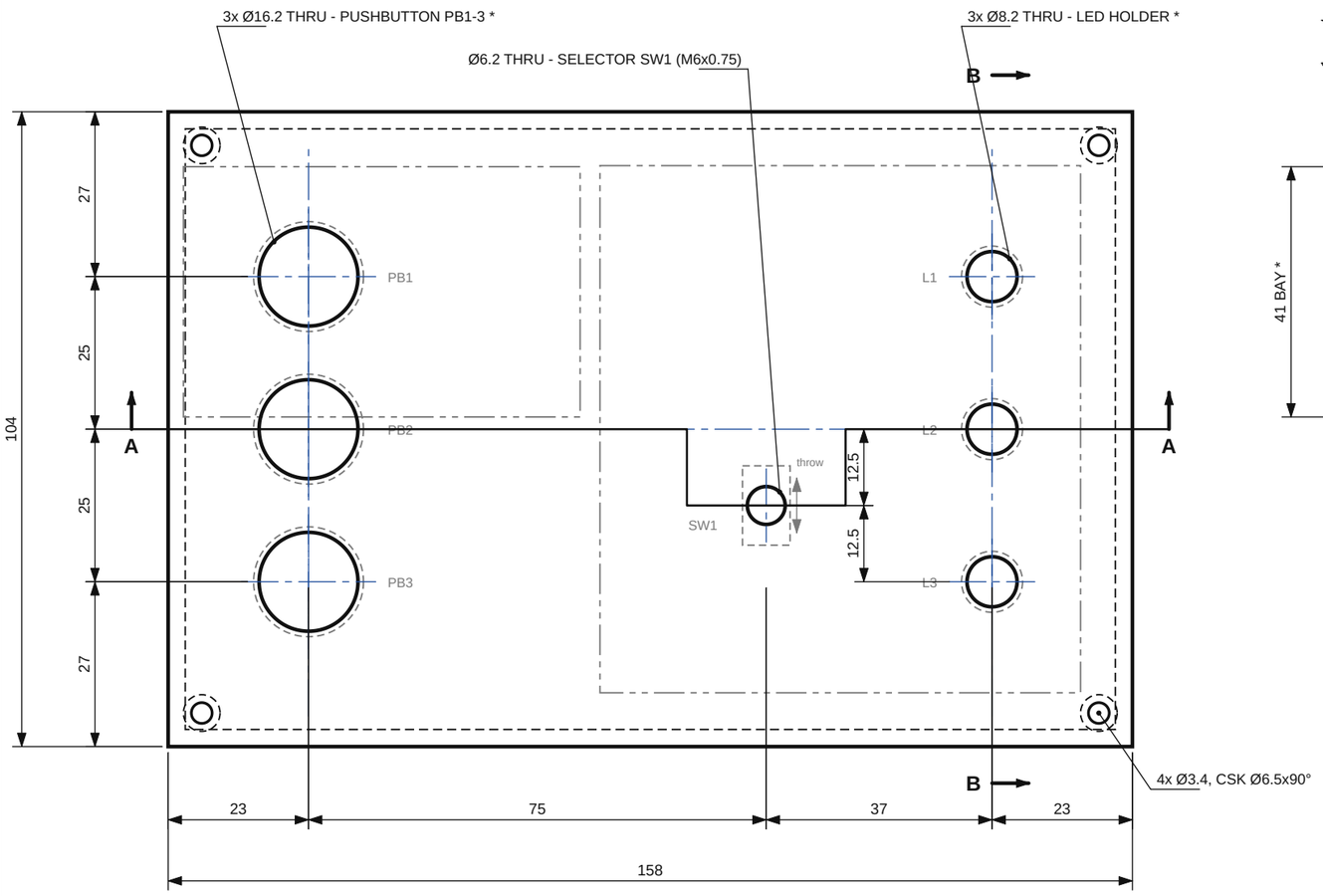
VIEW 1 - TOP (CONTROL PANEL / LID)

Lid 3.0 thick. Hidden: lid locating lip. Phantom: bays below.



VIEW 2 - PLAN, LID REMOVED

Holder long side left-right (photo), back-left | electronics RIGHT under LEDs

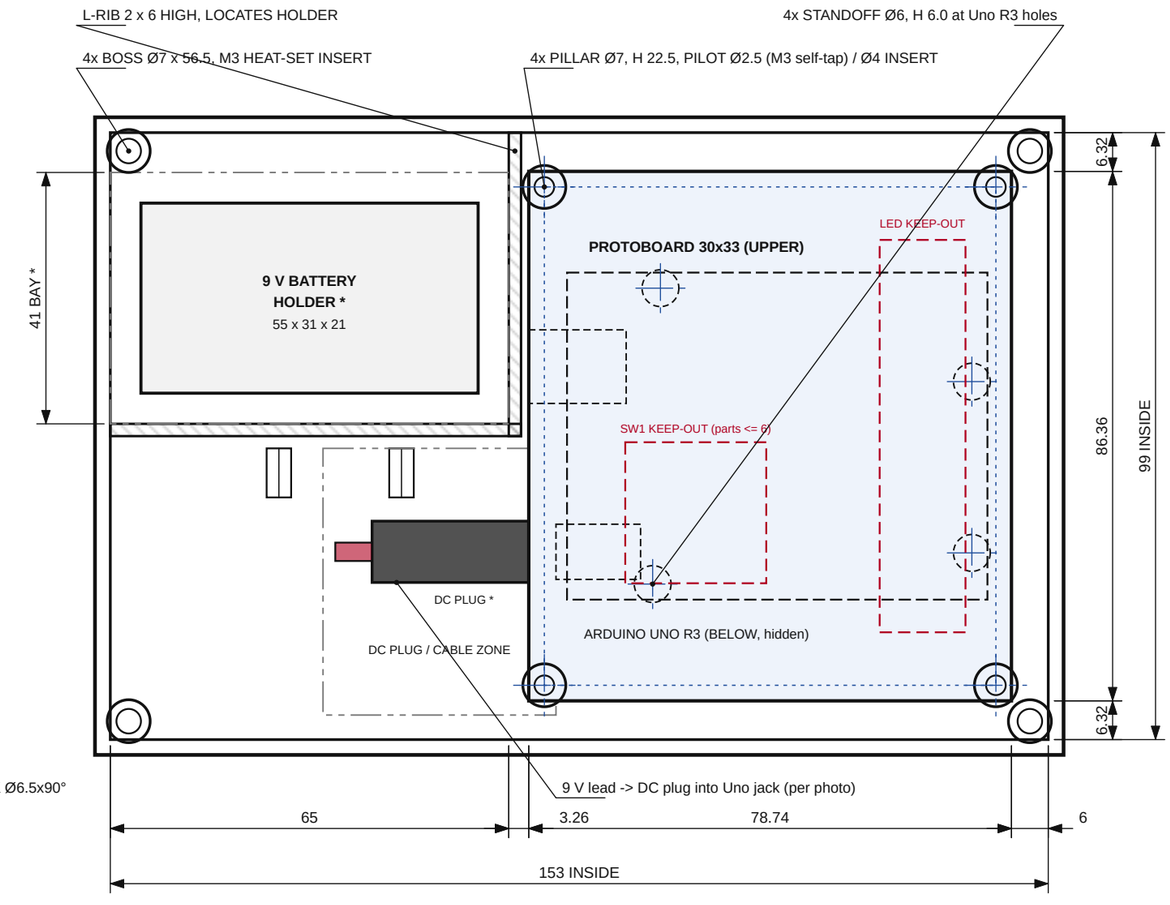


TABLE 1 - FEATURE COORDINATES (origin: external front-left-bottom corner, mm)

ITEM	X	Y	SIZE	NOTE
PB1/PB2/PB3	23	77 / 52 / 27	Ø16.2	Adafruit 1439, 16 mm illum. momentary *
SW1 selector	98	39.5	Ø6.2	Miyama MS-500E (ON)-OFF-(ON), M6x0.75
L1/L2/L3	135	77 / 52 / 27	Ø8.2	LED + 8 mm panel holder (model TBC) *
Lid screws	5.5 / 152.5	5.5 / 98.5	Ø3.4	4x M3 x 10 countersunk into inserts
Battery holder	7.5-62.5	59-90	55x31x21	flat, long side along X (photo) *
Protoboard	70.76-149.5	8.82-95.18	78.74x86.36	underside Z 25 *
PB pillars	73.3 / 146.96	11.36 / 92.64	Ø7 x 22.5	corner holes drilled to Ø3.2
Arduino Uno R3	77-145.58	25.33-78.67	68.58x53.34	underside Z 8.5, USB/DC jack face LEFT
Uno standoffs	90.97 / 92.24 / 143.04 / 143.04	27.87 / 76.13 / 32.95 / 60.89	Ø6 x 6.0	4 holes, ref. Uno R3 drawing

** = nominal / assumed value - see Sheet 3, Table 4. Uno standoff X and Y are listed pairwise in the same order.*

LINE LEGEND

- Visible outline
- Hidden edge
- Centre line
- Zone / phantom (under lid)
- Keep-out zone
- Cut material (sections)
- Protoboard (upper level)

KEY DECISIONS (spec §1)

- LEFT: 9 V holder, long side left-right (photo)
- RIGHT: protoboard above Uno R3, under LEDs
- SW1 right of centre, between L2/L3 rows (v6)
- Uno jacks face battery (photo); no USB hole
- Envelope: 158 x 104 x 62 (was 200x120x75)

REVISION HISTORY

A 2026-11-09 First issue from spec (9 Nov 2026).
B 2026-11-09 SW1 to layout-v6 position.
C 2026-11-09 Uno + 9 V plug per photo.
D 2026-11-09 Holder long side L-R (photo). * = assumed.

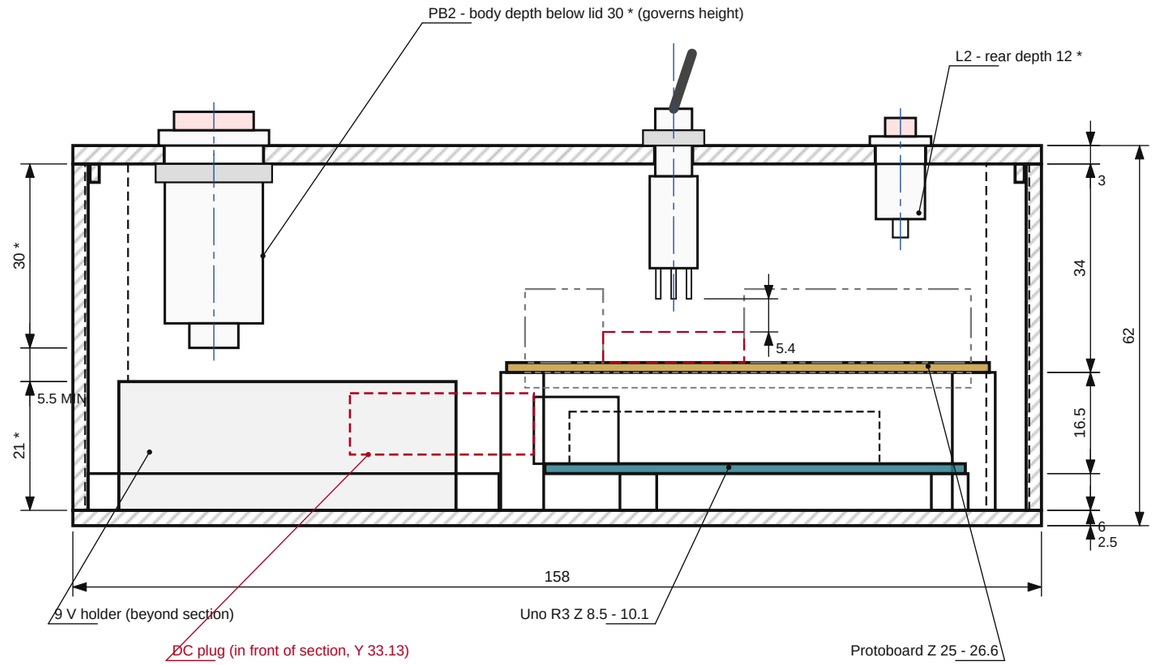
Drawn: Claude Code Checked: _____

Approved: _____

PROJECT: POWER GRID BOX - 3D PRINTED ENCLOSURE			DRAWING No.: PGB-ENC-001
TITLE: GENERAL ARRANGEMENT - PLAN VIEWS			SHEET: 1 OF 3
UNITS / SCALE: mm 1:1 @ A3	REV: D - PRELIM.	DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated	STATUS: DESIGN BRIEF - NOT FOR PRODUCTION	

SECTION A-A (stepped, viewed from front)

Stepped section: Y = 52 (PB2, L2), jog to Y = 39.5 through SW1



SECTION B-B (X = 135, viewed from right)

Through LED column L1-L3 and the board stack

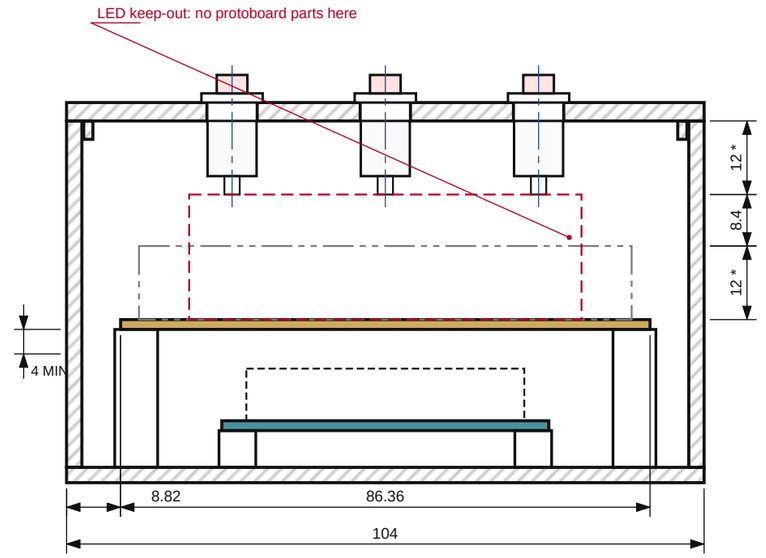


TABLE 2 - VERTICAL STACK-UP (Z from outside bottom)

LAYER	Z RANGE	NOTE
Floor	0 - 2.5	2.5 floor, 4x Ø10 x 1 foot recesses
Uno standoffs	2.5 - 8.5	6.0 high, Ø6
Uno R3 PCB	8.5 - 10.1	1.6 PCB
Uno parts / headers	10.1 - 21.0	DC jack/USB 10.9, headers 8.5
Clearance	21.0 - 25	4 to Uno parts, 3.4 to DC plug
Protoboard	25 - 26.6	on 4 floor pillars, H 22.5
Protoboard parts	26.6 - 38.6	max 12 * (keep-out under LEDs)
LED rear	47 - 59	12 *; 8.4 clear to parts
Battery holder	2.5 - 23.5	21 * high, on floor
Selector rear	37 - 59	22 *; parts under SW1 <= 5
Button rear	29.0 - 59.0	30 *; 5.5 clear to holder
Base top / lid	59.0 - 62.0	3.0 lid, 1.5x3 lip, 0.3 gap

Overall height is set by: holder 21 + 5.5 gap + button 30 + lid 3 + floor 2.5 = 62.

Each 1 mm saved on the measured button depth reduces height by 1 mm.

LEGEND

- Visible outline
- Hidden edge
- Centre line
- Zone / phantom (under lid)
- Keep-out zone
- Cut material (wall, floor, lid)
- 9 V battery holder
- Protoboard
- Arduino Uno R3

REVISION HISTORY

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B 2026-11-09 SW1 to layout-v6 position.
C 2026-11-09 Uno + 9 V plug per photo.
D 2026-11-09 Holder long side L-R (photo). * = assumed.

Drawn: Claude Code Checked: _____

Approved: _____

PROJECT: POWER GRID BOX - 3D PRINTED ENCLOSURE			DRAWING No.: PGB-ENC-001
TITLE: SECTIONS A-A / B-B, VERTICAL STACK-UP			SHEET: 2 OF 3
UNITS / SCALE: mm 1:1 @ A3	REV: D - PRELIM.	DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated	STATUS: DESIGN BRIEF - NOT FOR PRODUCTION	

GENERAL NOTES

1. All dimensions in mm. Third-angle projection. Model origin: external front-left-bottom corner (X right, Y back, Z up).
2. Two printed parts: BASE (tub with floor, walls, rib, bosses, pillars, standoffs) and LID (control panel). Print lid face-down.
3. Walls 2.5, floor 2.5, lid 3.0. External corners R3 (vertical edges) optional; internal fillets R1 at wall/floor joints.
4. Lid fixing: 4x M3 x 10 countersunk screws into M3 x 5.7 heat-set inserts (Ø4.0 pilot, 6 deep) in corner bosses Ø7.
5. Lid lip 1.5 wide x 3.0 deep, 0.3 clearance per side to inner wall; notch lip 8 x 8 around each corner boss.
6. Panel holes are printed at +0.2 over nominal (Ø16.2 / Ø6.2 / Ø8.2). Ream to fit after a test print of a 30 x 30 coupon.
7. SW1 anti-rotation: add a Ø2.5 x 1.5 deep key hole at 8 mm from SW1 centre once the switch's tab washer is confirmed.
8. Protoboard on 4 floor pillars (Ø7 x 22.5) at its corner hole positions, drilled to Ø3.2; M3 self-tapping or insert.
9. Uno R3 on 4 standoffs Ø6 x 6.0 (Ø2.5 pilot for M3 self-tap / nylon M3). USB/DC jack face LEFT (photo).
10. Battery service: remove 4 lid screws, fold lid over to the LEFT on a 150 mm wire service loop; holder is directly accessible. The protoboard / Uno stack is NOT disturbed (spec §6).
11. Wiring: the 9 V lead leaves the holder, passes over the 6 mm locating rib, runs along the floor in front of the bay (2 tie anchors at Y 46) into the DC plug, which goes straight into the Uno DC jack at Y 33.13.
12. Power per photo: 9 V holder lead ends in a DC barrel plug in the Uno jack. Keep the zone X 37.2-75.2, Y 2.5-52 free for the plug and its cable bend. No USB opening (spec §1). Fit the plug before mounting the protoboard.
13. Protoboard keep-outs: LED column (X 128-142, Y 20-84) parts <= 3; SW1 (X 86.5-109.5, Y 28-51) parts <= 5.
14. Add 4x Ø10 x 1.0 deep recesses on the underside at (12,12) (146,12) (12,92) (146,92) for rubber feet.
15. SW1 position follows the top layout of draft v6 (position only; no v6 dimensions used): right of centre, midway between the L2 and L3 rows. Mount the toggle so the lever throws along Y (back = towards L2, front = towards L3).

TABLE 3 - CLEARANCE CHECKS (computed from model)

CHECK	ACTUAL	MIN	RESULT
Button body -> battery holder (Z)	5.5	5	OK
LED rear -> protoboard parts (Z)	8.4	5	OK
Uno tallest part -> protoboard (Z)	4	2.5	OK
Protoboard edge -> right wall (X)	6	5	OK
DC plug -> battery bay rib (Y)	13.87	5	OK
Uno USB tip -> battery bay rib (X)	1.15	1	OK
Protoboard edge -> holder (X)	8.26	5	OK
DC plug top -> protoboard (Z)	3.4	2.5	OK
Protoboard edge -> front/back wall (Y)	6.32	5	OK
Uno edge -> right wall (X)	9.92	3	OK
Selector rear -> protoboard parts (Z)	5.4	5	OK
Button bezel -> bezel (Y)	7	5	OK
Corner boss -> protoboard corner	0.97	0.5	OK

OPEN ITEMS (spec §1, §7)

- External DC input: OPEN, not modelled. No USB opening.
- Confirm button quantity (3) and LED model / quantity (3).
- Confirm 2.8 x 0.5 QC receptacle fit on button blades.
- Verify SW1 terminal arrangement with continuity meter.
- Spec §5 says snap connector; photo shows a DC barrel plug. Drawn per photo.

A right-angle DC plug would cut about 20 mm from the width.

TABLE 4 - ASSUMPTIONS REGISTER (* values: the spec lists these as TBC/approx. - measure with calipers and update the parameters)

ITEM	VALUE USED	BASIS / SOURCE	ACTION
Protoboard outline	78.74 x 86.36 x 1.6	Hole span 73.66 x 81.28 (spec §4) + 2.54 edge margin each side	Measure edge-to-edge
Protoboard part height	12 max above board	Typical radial electrolytic / terminal block	Measure tallest part
Arduino Uno R3	68.58 x 53.34 x 1.6	Arduino reference design; holes (13.97,2.54) (15.24,50.80) (66.04,7.62) (66.04,35.56)	Verify on board
Uno part height	10.9 (USB-B / DC jack)	Reference design; headers 8.5	Verify
9 V battery holder	55 x 31 x 21, flat, long side L-R	Spec §4 'approx. 55 x 31 x 21' - not verified	Measure + wire exit
Battery bay reserve	65 (X) x 41 (Y)	Spec §4 provisional planning reserve	Re-check after holder
Adafruit 1439 button	Ø16 mount, Ø18 bezel, 30 below panel	16 mm nominal (spec); depth incl. 2.8 mm QC terminal + wire bend ASSUMED	CRITICAL: sets height
Miyama MS-500E	Ø6.2 bore, body 7.8 x 13, 22 below panel	Spec §4 listing; depth below panel ASSUMED	Check mfr. drawing
LED indicator	Ø8.2 cut-out, 12 rear depth	Generic 5 mm LED in 8 mm panel holder - model not chosen	Select model
Panel thickness	3.0 lid	Within typical max panel thickness for M6 bushing / 16 mm button	Check both parts
DC plug (9 V lead)	30 long, Ø10, +8 bend	Arrangement from user photo; size estimated - photo not used as a measurement	Measure plug + boot
Clearance allowance	>= 5 (6 at electronics bay)	Spec §6 preliminary allowance; 6 used so corner bosses clear the board	-

HOW TO UPDATE AFTER MEASURING

a) Edit the parameter block at the top of tools/gen_drawing.py with the measured values.
b) Overall size follows automatically: width = wall + battery bay (65) + rib + protoboard + 6 + wall;
depth = walls + 2 x 6 + protoboard depth; height = floor + holder + 5 + button depth + lid (re-check Table 3).
c) Re-run tools/gen_drawing.py - the clearance table (Table 3) and the asserts fail if any check drops below its minimum.
d) Before the full print: print the lid only (approx. 1.5 h) and trial-fit PB1-3, SW1 and L1-3.

REVISION HISTORY

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B 2026-11-09 SW1 to layout-v6 position.
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D 2026-11-09 Holder long side L-R (photo). * = assumed.

Drawn: Claude Code Checked: _____

Approved: _____

PROJECT: POWER GRID BOX - 3D PRINTED ENCLOSURE			DRAWING No.: PGB-ENC-001
TITLE: NOTES, ASSUMPTIONS REGISTER, VERIFICATION			SHEET: 3 OF 3
UNITS / SCALE: mm 1:1 @ A3	REV: D - PRELIM.	DATE: 2026-11-09	PROJECTION
MATERIAL / PROCESS: PETG, FDM, 0.2 mm layers	GENERAL TOL.: +/-0.3 unless stated	STATUS: DESIGN BRIEF - NOT FOR PRODUCTION	